Code and workflow: github.com/YangzhenyuGao/BIOS774_Agent3_RepresentationLearning
Study reports: Study 1: PBMC RNA + ATAC · Study 2: TCGA-BRCA histology + clinical

An Auditable Agent for Multimodal Biomedical Representation Learning

Yangzhenyu Gao · BIOS 774 Project 3

Abstract

We present a rule-based agent that profiles paired modalities, chooses preprocessing and learning methods, runs validation pilots, freezes models and generates source-linked reports. For PBMC RNA–ATAC, contrastive learning improves retrieval but reduces clustering agreement and adds no convincing accessibility-prediction benefit. For breast histology and clinical data, a latent probe reaches macro-F1 0.723, but gains depend on receptor covariates and the native head is weak. The experiments demonstrate reproducible automation while exposing incomplete objective enforcement, inadequate convergence auditing and narrow foundation-model comparisons. They do not establish generally superior multimodal representations.

1 System design and agent decisions

The task is to decide *how* modalities should be integrated, rather than replay a fixed pipeline. Typed data and objective specifications encode modality type, sparsity, pairing unit, permitted supervision, downstream tasks and forbidden inputs. A deterministic registry supplies genuine adapters; validation evidence selects a shortlist, and saved test evidence supports a conditional integration verdict. The reproducible core uses rules and Jinja2/WeasyPrint reports, not an external LLM.

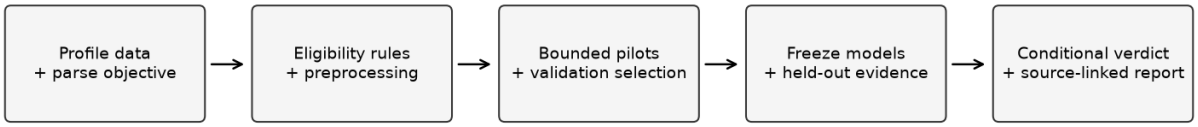


Figure 1. One orchestration graph serves both applications. Content hashes, fit-scope records and failure receipts accompany every stage. Auditability does not imply that the implemented acceptance criteria fully capture the objective.

RNA + ATAC. Exactly paired PBMC cells from one healthy donor [1] trigger training-fitted RNA normalization/HVG/PCA and ATAC TF-IDF/LSI; MultiVI receives raw expression counts and binary accessibility. WNN, MultiVI, Gaussian MOFA+, self-supervised feature-dropout learning and paired InfoNCE are eligible [2,5–8]. QC retains 10243 cells (7171/1532/1540 train/validation/test). FixFlip cell types fit downstream probes, never the primary representation; they derive from WNN annotations [14].

Histology + clinical. TCGA-BRCA diagnostic slides are grouped by patient [3,4]. Tissue/QC filtering and physical-resolution tiling precede frozen patch encoders and attention MIL. Training-fitted clinical inputs are age, stage, histology and ER/PR/HER2 status, with missingness indicators. Four-class RNA-derived PAM50 is the target; molecular target fields, outcomes and treatment variables are excluded from inputs. Labels may train supervised fusion; image–table contrastive learning and actual local CONCH-Z tower tuning are alternative arms [9–13]. The cohort contains 1132 primary slides from 1061 patients, with 802 four-class labels.

2 Selection and evaluation protocol

Patient/cell splits precede fitted transformations; every patient's slides remain together. Small validation grids tune the same standardized logistic and kNN probes across representations. An epsilon-Pareto rule over declared primary metrics, practical-equivalence filtering and simplicity select at most two candidates for tuning. Final models are frozen before held-out evaluation. Classification, clustering, alignment, masking, stability and technical confounding remain separate evidence categories; paired bootstrap intervals quantify test differences.

What this guarantees. The protocol controls leakage and makes failures traceable. It does not guarantee sufficient optimization, equal effective data access or correct translation of a prose objective into hard acceptance criteria. The results below distinguish these limitations from successful execution.

3 PBMC: alignment improves; cell-type structure does not

Representation	Validation F1	Val. R@10	Test F1	ARI	NMI
RNA PCA	0.902	0.439	0.909	0.613	0.756
ATAC LSI	0.808	0.439	0.853	0.576	0.717
Scaled concatenation	0.925	0.439	0.910	0.679	0.807
Inductive WNN	0.904	0.263	0.886	0.451	0.724
MultiVI	0.716 ± 0.042	0.133	—	—	—
Gaussian MOFA+	0.887 ± 0.000	0.395	—	—	—
SimCLR-style	0.925 ± 0.002	0.342	—	—	—
Paired contrastive	**0.916 ± 0.004**	**0.510**	**0.925**	**0.528**	**0.777**

Table 1. Complete PBMC pilot comparison and available frozen test results. Validation is mean ± population SD over three declared seeds for learned candidates; reference runs are single fits. Tests use the predeclared representative seed and common logistic probe. "—" means not advanced to final evaluation, not a measured poor test result. Three PCA/LSI reference retrieval entries share one ridge mapping.

3.1 Objective satisfaction differs from selection

The contrastive model traded slightly lower validation F1 within the predeclared equivalence margin for better retrieval. Test symmetric paired recall@10 rises from 0.428 for the common PCA→LSI ridge reference to 0.497: difference 0.069, 95% CI [0.050, 0.090]. Macro-F1 gains remain uncertain: 0.016 versus RNA alone (95% CI [-0.009, 0.060]) and 0.015 versus concatenation ([-0.007, 0.059]). Leiden ARI is **0.528**, below RNA PCA (**0.613**) and concatenation (**0.679**). NMI is mixed (Table 1), improving over RNA alone but remaining below concatenation; it does not resolve the ARI regression.

The objective requested structure at least as good as the better single modality. The selector made classification F1 and retrieval primary, leaving ARI secondary; it did not enforce structural noninferiority. The full PBMC objective was therefore **not met**, despite successful selection and a frozen automated "integrate with caution" flag. This is an objective-to-criterion translation failure.

3.2 Expression-to-accessibility prediction adds little

RNA→ATAC predictor	Cell AUROC	Cell AUPRC	Peak AUPRC
Training peak frequency	0.779	0.511	0.211
RNA PCA + kNN decoder	0.800	0.506	0.284
Paired contrastive + RNA decoder	0.801	0.506	0.286

Table 2. Existing held-out predictions; target ATAC input is masked. Cell and peak AUPRC average different ranking problems.

The learned RNA tower barely changes prediction relative to RNA PCA, while training peak frequency has the highest per-cell AUPRC. A small per-peak AUPRC increase does not establish useful incremental prediction; no uncertainty estimate was saved for that increase. The row originally labeled as the ATAC-only model's decoder actually calls its *RNA-side* encoder. RNA-only, ATAC-named and concatenation decoder rows are identical to RNA PCA and are collapsed here. They are neither independent evidence nor ATAC-only predictors. Their retrieval scores also reuse one training-fitted ridge map; this report counts **one alignment reference**, not three corroborating comparisons.

3.3 Weak pilots do not establish inferior methods

MultiVI was capped at 50 epochs with 25 epochs of KL warmup. Worker logs reached the cap, but saved curves are empty, preventing a convergence audit. Framework-default early stopping was disabled, although a custom validation callback exists. The API default is 500 epochs [15]: this is a short pilot, not a fair converged-method ranking. WNN uses a muon graph, diffusion coordinates and kNN out-of-sample interpolation. Training top-15 graph overlap with Seurat WNN is only 0.195. Compression, graph differences and extrapolation can weaken this adapter; its F1 does not show canonical WNN is inferior. WNN-derived expert labels would otherwise favor a closely matched native analysis, but do not guarantee that this different inductive embedding preserves their structure.

4 TCGA: receptor-dependent utility and a weak native head

Candidate / feature source	Validation macro-F1	Frozen test F1
Frozen UNI2-h mean	0.665	0.579
Frozen CONCH-Z mean	0.605	0.570
ImageNet ResNet-18 mean	0.535	0.421
Clinical only	0.568	0.620
Scaled concatenation	0.720	0.666
Attention MIL	0.696 ± 0.012	0.645
SimCLR-style	0.692 ± 0.010	—
Feature-pair contrastive	0.711 ± 0.031	—
Early fusion	0.729 ± 0.020	—
Gated fusion	0.723 ± 0.014	—
Late fusion	**0.759 ± 0.012**	**0.723**
CONCH-Z last-block tuning	0.593 ± 0.026	—
CONCH-Z image–text tuning	0.453 ± 0.006	—

Table 3. All executed TCGA pilot strategies, including both foundation-model tuning arms. Mean ± SD is over three seeds for neural methods; frozen mean/clinical/concatenation references use one fit. MIL, fusion, feature-pair contrastive and SimCLR use UNI2-h features; tower tuning uses CONCH-Z. Test F1 is always a common logistic-probe score. Unselected candidates were not opened on the test set.

4.1 Receptor dependence limits the integration claim

The selected representation concatenates attention-MIL image and clinical-MLP latents. Its probe F1 is 0.723, against 0.645 for image-only MIL and 0.666 for concatenation. Paired differences remain uncertain: 0.078 versus MIL (95% CI [-0.009, 0.170]) and 0.056 versus concatenation ([-0.035, 0.141]).

Existing same-test-set comparison	Macro-F1	Her2 recall
Image-only attention MIL + probe	0.645	—
Fusion + probe, all clinical variables	0.723	0.700
Fusion refit + probe, ER/PR/HER2 removed	0.583	—
Fusion native classification head	0.482	0.000

Table 4. Receptor removal refits the same family and probe on the original split; it is not merely test-time masking. Representation quality, receptor sensitivity and native-head behavior are distinct outcomes.

Without ER/PR/HER2, fusion falls to 0.583, **below same-test image-only MIL**. A receptor-independent integration benefit is not supported. Receptors are biologically related PAM50 proxies and allowed clinical inputs, not prohibited molecular target copies. Their contribution nevertheless qualifies task interpretation. This ablation shows dependence in this system, not a causal decomposition of every gain; a receptor-only control and a paired interval for this specific comparison were not run.

4.2 The probe is not the trained classification head

Training averages the branches' class logits under unweighted cross-entropy. The native head reaches only 0.482 F1 and recalls none of 10 test Her2 cases. Her2 is a minority class (65 of 802 labeled cohort patients), so imbalance and branch-logit averaging are plausible contributors; their causal roles cannot be isolated from these runs.

The probe fits a new standardized, regularized linear boundary on concatenated frozen latents, selecting regularization on validation. It reaches 0.723 F1 and 0.700 Her2 recall, showing that the latent retains information the native decision rule fails to use. **The probe is also unweighted**; weighting does not explain its advantage. The native head is inadequate as a four-subtype classifier and must not be described using the probe's score.

5 What the agent learned—and failed to establish

5.1 Alignment is not a uniformly mixed biological manifold

The paired towers show a visible modality gap despite better retrieval. Ranking a cell's partner and merging both modality populations into a common manifold are different requirements. The display does not prove complete separation, but it does not justify claiming full mixing. Together with lower ARI and weak accessibility gains, it restricts the PBMC conclusion to alignment under the chosen retrieval criterion. Technical depth/QC remains predictable; one donor cannot establish donor transfer.

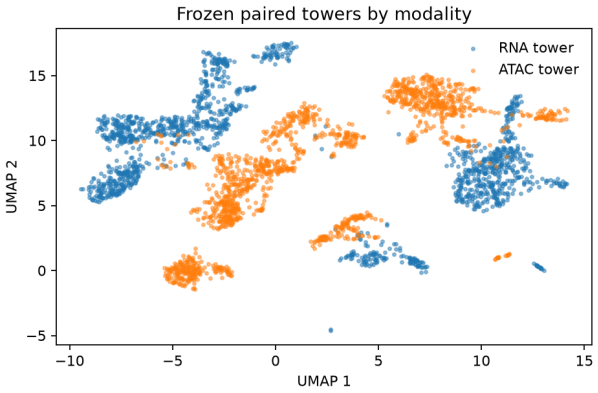


Figure 2. Saved RNA/ATAC tower display: fit training embeddings, transform test cells. Visible offsets remain; UMAP is descriptive, not a new test or selection criterion.

For TCGA, source-site predictability remains a concern. A separately refitted site-held-out analysis reaches F1 0.679; its different test population prevents a paired comparison with Table 4. Exploratory survival C-index is 0.665 (95% CI [0.558, 0.770]; 30 test deaths), which does not establish prospective benefit.

5.2 Foundation-model and resource choices were conservative, not optimized

UNI2-h and CONCH-Z enter production comparisons alongside ImageNet. Virchow2 and MUSK only passed loading/runtime checks, without empirical comparison. Encoder choice reflects compatibility, feature size and compute defaults, not validation evidence that these encoders are best. Public-model pretraining overlap with TCGA cannot be ruled out.

The cap of 128 tiles per slide arose from two short-queue shards and a two-hour task limit. This was an operational choice, not an aggregate-budget necessity: the saved projection for 256 tiles is 7.881 GPU-hours, while project usage was 2.770 of 48 authorized GPU-hours. More sequential shards could retain the concurrency limit while covering more tiles. No larger-cap performance was measured, so its benefit remains unknown.

CONCH-Z tuning genuinely updated pretrained parameters, but used only 128 training patients and four cached tiles each; inference used the first single cached tile. Its low scores compare unequal effective data access and capacity, not the best achievable tuned foundation model. These arms demonstrate execution capability, not competitive fine-tuning or superiority of frozen encoders.

5.3 Automation strengths, failure modes and next design changes

The agent orchestrates genuine adapters, leakage-controlled splits, validated resumes, failures, frozen representations and source-linked reports across different pairing structures. Synthetic fixtures exercise serialization, true/shuffled pairing, a third modality configuration and objective-sensitive planning. These software achievements are distinct from evidence of useful multimodal synergy.

The audit exposes decision-layer weaknesses: structural noninferiority was not enforced; reused references appeared independent; a decoder label obscured its input; empty histories did not block interpretation; and resource settings were not revisited despite unused budget. A future agent should require objective-specific acceptance tests, explicit baseline identity, convergence evidence, class-wise head diagnostics and comparable training data access before recommending integration. These are proposed changes, *not evaluated repairs*.

6 Conclusion

> **PBMC:** retrieval improves, but structure-preservation and incremental RNA→ATAC prediction goals are not established. **TCGA:** useful frozen-latent probe performance is conditional on receptors, without clear general integration benefit; the native head fails a minority subtype. The unchanged automated flags say “integrate with caution.” This stricter report-level audit finds that neither experiment warrants an unqualified claim of successful multimodal representation learning.

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Appendix A Objective-dependent planning checks

These saved checks change the objective while retaining the same data profile. No alternative model is trained and no alternative held-out performance is claimed. Survival remains secondary in the executed TCGA experiment.

PBMC: evaluation-only to label-informed planning

Candidate	Original eligibility / loss	Changed eligibility / loss
Early fusion	reference only cross entropy	eligible for pilot cross entropy
Late fusion	reference only cross entropy	eligible for pilot cross entropy
Gated fusion	reference only cross entropy	eligible for pilot cross entropy

TCGA: subtype classification to survival planning

Candidate	Original eligibility / loss	Changed eligibility / loss
Early fusion	eligible for pilot cross entropy	eligible for pilot Cox partial likelihood
Late fusion	eligible for pilot cross entropy	eligible for pilot Cox partial likelihood
Gated fusion	eligible for pilot cross entropy	eligible for pilot Cox partial likelihood
Attention MIL	eligible for pilot cross entropy	eligible for pilot Cox partial likelihood
CONCH-Z last-block tuning	eligible for pilot cross entropy	eligible for pilot Cox partial likelihood

The PBMC swap changes supervised-fusion eligibility and exchanges retrieval for balanced accuracy as a primary metric. The TCGA swap changes survival-capable candidates to Cox partial-likelihood and concordance-based planning. This demonstrates objective-sensitive rules, not correct fulfillment of every objective constraint; Section 3 shows the structural failure.

Reproducibility and revision boundary

This revision reads saved results, existing source and worker logs. It does not retrain, retune, recompute held-out metrics, generate new UMAP coordinates or change machine verdicts. Original generated dataset reports are retained. A pre-revision report is preserved locally; current numeric claims map to saved artifacts in the claim-source file.

Source, objectives, configurations, environment locks and manifests identify the executed workflow. Shared raw data, tile features, checkpoints and pathology weights/derivatives are not redistributed. All downstream files remain inside Agent3. No external LLM was used in the reproducible analysis or templated reporting core.

Appendix B Audit trail and interpretation corrections

Finding	Evidence inspected	Interpretive correction
Receptor dependence	Existing receptor-removal refit and image-only MIL probe	Place receptor-free fusion beside MIL; no independent benefit is established.
Native-head failure	Native-head / probe test records, neural loss and probe fitting	Different classifiers; both unweighted. Minority-class failure remains explicit.
MultiVI optimization	Three worker logs, empty curves, custom validation callback	Short pilot; adequate convergence unverified. Do not infer method inferiority.
Noncanonical WNN	Muon graph, diffusion coordinates, interpolation and R cross-check	Adapter performance does not rank canonical Seurat WNN.
Objective mismatch	Objective, selector, clustering evidence and modality display	Retrieval improves while structure deteriorates; full objective unmet.
Decoder label	Baseline encoder's requested modality and RNA→ATAC caller	The ATAC-named decoder uses RNA. Collapse duplicates.
Repeated retrieval reference	Identical saved alignment records for three feature references	One ridge map, not multiple independent baselines.
Foundation-model scope	Encoder selection, throughput and bounded tuning code	Two production pathology encoders; sparse coverage and unequal data access.

Machine-readable sources

Experimental summaries: `outputs/<dataset>/evidence_bundle/pilot_summary.json`; `evaluation/evidence.json`; `downstream/results.json`; `verdict.json`. Resource evidence: TCGA `features/encoder_selection.json` and `logs/gpu_ledger.json`. Code/log audit hashes: `reports/report_audit_context.json`. Per-number provenance: `reports/report_claim_sources.json`. Page/source acceptance: `reports/project_report_validation.json`. Objective swaps: `outputs/robustness/objective_swap_summary.json`.

Validation and test columns remain distinct; missing test results are never imputed from pilots. The report's stricter interpretation is an editorial assessment of frozen evidence, not a changed selection rule, new hypothesis test or post-hoc model search.